

CIVIL NUCLEAR ENERGY AND INTERNATIONAL LAW: FRAMEWORKS, OBLIGATIONS, AND EMERGING CHALLENGES

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Abstract

The peaceful use of nuclear energy is one of the most consequential intersections between science, sovereignty, and international law. This article examines the normative architecture governing civil nuclear energy at the international level, tracing the evolution of legal obligations from the foundational Nuclear Non-Proliferation Treaty (NPT) of 1968 through the work of the International Atomic Energy Agency (IAEA) and beyond. It analyses the right to peaceful use of nuclear energy, the principle of non-proliferation, the role of safeguards agreements, liability regimes for nuclear damage, and the emerging legal questions surrounding nuclear security, radioactive waste management, and climate-driven nuclear expansion. The article also addresses the relevance of these frameworks to emerging nuclear states, with particular reference to India's unique legal position outside the NPT. It concludes by identifying areas requiring progressive development of international law to meet the challenges of the twenty-first century.

Keywords: *Nuclear Non-Proliferation Treaty; IAEA Safeguards; Peaceful Use of Nuclear Energy; Vienna Convention on Nuclear Liability; Nuclear Security; India-US Civil Nuclear Deal; International Nuclear Law*

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INTRODUCTION

Nuclear energy occupies a paradoxical space in the international legal order. Born from the weapons programme that devastated Hiroshima and Nagasaki in 1945, it was rapidly recast as a technology of development and peace through President Eisenhower's landmark 'Atoms for Peace' address before the United Nations General Assembly in December 1953. That address heralded the creation of the International Atomic Energy Agency (IAEA) in 1957 and laid the groundwork for a comprehensive international legal regime governing the civilian use of atomic power.

Today, nuclear energy supplies approximately ten per cent of global electricity generation and is increasingly positioned as a low-carbon option in the transition away from fossil fuels. As of 2024, thirty-two countries operate nuclear power plants, with a further twenty countries actively pursuing civil nuclear programmes. Against this backdrop, international law governing civil nuclear energy has grown from a sparse set of treaty commitments into a multi-layered regime encompassing non-proliferation obligations, safeguards, civil liability, physical protection, radioactive waste management, and emergency preparedness.

This article provides a comprehensive survey of the international legal framework applicable to civil nuclear energy, identifies its principal normative pillars, and examines tensions and lacunae that challenge its continued efficacy. Part II traces the foundational architecture. Part III analyses safeguards and verification. Part IV examines nuclear liability law. Part V addresses nuclear security. Part VI considers radioactive waste and decommissioning. Part VII discusses India's exceptional legal position. Part VIII identifies emerging challenges before concluding observations in Part IX.

THE FOUNDATIONAL LEGAL ARCHITECTURE

A. The Nuclear Non-Proliferation Treaty, 1968

The Treaty on the Non-Proliferation of Nuclear Weapons (NPT), which entered into force on 5 March 1970, constitutes the cornerstone of the international non-proliferation regime. Its architecture rests upon three mutually reinforcing pillars: non-proliferation, disarmament, and the inalienable right to the peaceful use of nuclear energy.

Article I obliges nuclear-weapon States (NWS) - defined in the Treaty as those that manufactured and exploded a nuclear device before 1 January 1967, namely the United States, Russia, the United Kingdom, France, and China - not to transfer nuclear weapons or control

over them to any recipient, nor to assist non-nuclear-weapon States (NNWS) in acquiring such weapons. Article II obliges NNWS not to receive, manufacture, or otherwise acquire nuclear weapons.¹

Crucially, Article IV preserves the inalienable right of all States Parties to develop research, production, and use of nuclear energy for peaceful purposes.² This provision has been the subject of sustained interpretive controversy: does the right to peaceful use extend to uranium enrichment and plutonium reprocessing - technologies that are inherently dual-use? The negotiating history and subsequent State practice suggest that Article IV encompasses the full fuel cycle, subject to safeguards obligations under Article III. However, the international community has increasingly sought to limit the spread of enrichment and reprocessing technology through supplier-group arrangements such as the Nuclear Suppliers Group (NSG).³

As at 2024, the NPT counts 191 States Parties, making it one of the most widely adhered-to arms control treaties in history. India, Pakistan, and Israel remain outside the Treaty, while North Korea purported to withdraw in 2003, though the legal validity of that withdrawal is contested.

B. The International Atomic Energy Agency

Established by the IAEA Statute in 1956, the Agency is mandated to accelerate and enlarge the contribution of atomic energy to peace, health, and prosperity throughout the world, while ensuring that its assistance is not used to further any military purpose.⁴ The IAEA performs three core functions: safety standards-setting, technical cooperation, and verification through safeguards.

The Agency's legal authority derives from a network of agreements including: (i) safeguards agreements with Member States under NPT Article III; (ii) the Additional Protocol (INFCIRC/540) which enhances inspection rights; (iii) project and technical cooperation agreements; and (iv) bilateral nuclear cooperation agreements.⁵ The IAEA Board of Governors and General Conference constitute its principal decision-making organs, while the Director General serves as the chief administrative officer.

¹ Treaty on the Non-Proliferation of Nuclear Weapons, 1 July 1968, 729 UNTS 161, Arts. I–II.

² Ibid, Art. IV.

³ Nuclear Suppliers Group, 'Guidelines for Nuclear Transfers' (INFCIRC/254/Rev.14/Part 1, IAEA, Vienna, 2022).

⁴ IAEA Statute, 26 October 1956, 276 UNTS 3, Art. II.

⁵ IAEA Statute (n 3) Art. III(A)(5); C. Stoiber et al., Handbook on Nuclear Law (IAEA, Vienna, 2003) 15-22.

The IAEA issues legally non-binding but practically authoritative Safety Standards in the form of Safety Fundamentals, Safety Requirements, and Safety Guides. These standards, developed through expert consensus, are incorporated into domestic law by most operating States and inform bilateral nuclear cooperation agreements globally.

Table 1: Key International Legal Instruments in Civil Nuclear Energy

Instrument	In Force	Principal Subject Matter	Parties
Treaty on the Non-Proliferation of Nuclear Weapons (NPT)	1970	Non-proliferation, disarmament, peaceful use	191
IAEA Statute	1957	Establishment of IAEA; safeguards mandate	176
Vienna Convention on Civil Liability for Nuclear Damage	1977/2003	Third-party nuclear liability	39
Paris Convention on Nuclear Third Party Liability	1968	Western European liability regime	16
Convention on Nuclear Safety	1994	Safety obligations for land-based reactors	92
Joint Convention on Safety of Spent Fuel	2001	Waste management obligations	87
Convention on Physical Protection (CPPNM/2005 Amend.)	2016	Physical security of nuclear material	163
CSA (INFCIRC/153)	Ongoing	State-level IAEA safeguards	182

Instrument	In Force	Principal Subject Matter	Parties
Additional Protocol (INFCIRC/540)	Ongoing	Enhanced verification rights	140+
Convention on Supplementary Compensation (CSC)	2015	Supplementary nuclear liability fund	12

C. Nuclear-Weapon-Free Zone Treaties

Regional Nuclear-Weapon-Free Zone (NWFZ) treaties complement the global NPT regime by prohibiting the stationing, testing, and use of nuclear weapons within defined geographic areas while explicitly affirming the right to peaceful nuclear energy. The Treaty of Tlatelolco (1968, Latin America and the Caribbean), the Treaty of Rarotonga (1986, South Pacific), the Treaty of Bangkok (1997, Southeast Asia), the Treaty of Pelindaba (2009, Africa), and the Treaty of Semipalatinsk (2009, Central Asia) collectively establish substantial NWFZ coverage across the Global South. These instruments create positive obligations for civil nuclear cooperation and negative security assurances from NWS Parties.

SAFEGUARDS AND VERIFICATION

A. The Comprehensive Safeguards Agreement

Article III of the NPT obliges each NNWS Party to conclude a Comprehensive Safeguards Agreement (CSA) with the IAEA. The model text for CSAs - INFCIRC/153 - requires States to declare all nuclear material subject to the Treaty and accept IAEA verification through on-site inspections, material accountancy, and containment and surveillance measures.⁶ The IAEA verifies the non-diversion of declared nuclear material to non-peaceful purposes.⁷

However, the CSA system was exposed as inadequate by Iraq's clandestine weapons programme discovered after the 1991 Gulf War. Iraq had declared its safeguarded facilities accurately but pursued covert enrichment outside the declared programme. The IAEA lacked authority under

⁶ IAEA, 'The Structure and Content of Agreements Between the Agency and States Required in Connection with the Treaty on the Non-Proliferation of Nuclear Weapons' (INFCIRC/153 (Corrected), IAEA, Vienna, 1972) para. 1.

⁷ Ibid, paras. 2-3.

the CSA to investigate undeclared activities. This vulnerability prompted negotiation of the Additional Protocol.

B. The Additional Protocol

The Model Additional Protocol (INFCIRC/540), approved by the IAEA Board of Governors in 1997, significantly expands the IAEA's verification authority. Key innovations include: (i) expanded declarations covering the entire nuclear fuel cycle, including uranium mining and fuel fabrication; (ii) short-notice complementary access to any location a State declares or the IAEA deems relevant; (iii) environmental sampling at and beyond declared sites; and (iv) collection of information on nuclear fuel cycle research and development.⁸ The Additional Protocol has become the de facto standard of verification adequacy, with many nuclear supplier States making its conclusion a condition for nuclear cooperation. As of 2024, over 140 States have the Additional Protocol in force.⁹

C. State-Level Concept and Integrated Safeguards

The IAEA has progressively developed a State-Level Concept (SLC) which evaluates the absence of undeclared nuclear material and activities in a State as a whole, rather than limiting analysis to declared facilities.¹⁰ States for which the IAEA has drawn a credible assurance of absence of undeclared material - the 'broader conclusion' - may be eligible for Integrated Safeguards (IS), which rationalises inspection frequency based on verified risk assessment. IS represents a sophisticated evolution of the verification system, moving from accountancy-based to intelligence-informed assurance.¹¹

INTERNATIONAL NUCLEAR LIABILITY LAW

A. The Vienna-Paris Framework

The international law of nuclear liability is governed by two principal instruments: the Vienna Convention on Civil Liability for Nuclear Damage (1963, as revised in 1997) and the Paris Convention on Third Party Liability in the Field of Nuclear Energy (1960, as revised). A Joint

⁸ IAEA, 'Model Protocol Additional to the Agreement(s) between State(s) and the International Atomic Energy Agency for the Application of Safeguards' (INFCIRC/540 (Corrected), IAEA, Vienna, 1997) Art. 5.

⁹ Ibid, Arts. 4–5.

¹⁰ IAEA, 'Safeguards Implementation Report for 2022' (GOV/2023/23, IAEA, Vienna, 2023) 7.

¹¹ IAEA, 'The Evolution of IAEA Safeguards' (International Verification Series No. 2, IAEA, Vienna, 1998) 18-24.

Protocol concluded in 1988 extends the benefits of each convention to nationals of the other, creating a degree of inter-system coherence.¹²¹³

Both instruments rest upon common principles: (i) absolute liability of the nuclear operator regardless of fault; (ii) exclusive liability, channelled to the licensed operator; (iii) limitation of liability in amount and time; and (iv) exclusive jurisdiction of the courts of the installation State.¹⁴ The 1997 Vienna Protocol substantially revised the original text, extending the geographical scope of compensable damage to include the exclusive economic zone, adding environmental damage and preventive measures as compensable heads, and raising the liability ceiling.

B. The Convention on Supplementary Compensation

The Convention on Supplementary Compensation for Nuclear Damage (CSC), adopted by the IAEA in 1997 and in force since 2015, creates a two-tier compensation system applicable regardless of whether the installation State is a party to the Vienna or Paris conventions. The first tier requires States to ensure a minimum national compensation fund. The second tier is a collectively-funded international pool drawn upon when first-tier funds are exhausted. The United States, Japan, India, Romania, and Morocco are among the Parties. The CSC is notable for its compatibility with national liability legislation that does not necessarily conform to the channelling model, making it more accessible to States with divergent domestic regimes.¹⁵

C. The Indian Civil Liability for Nuclear Damage Act, 2010

India's Civil Liability for Nuclear Damage Act, 2010 (CLNDA) generated significant international controversy upon its enactment. Section 17(b) of the CLNDA provides the operator a right of recourse against a supplier where the nuclear incident has resulted from a latent or patent defect in the supplied equipment or material,¹⁶ and Section 46 preserves the application of any other law in force.¹⁷ These provisions were widely criticised by foreign nuclear suppliers as incompatible with the exclusive liability and channelling principles of the CSC, to

¹² Vienna Convention on Civil Liability for Nuclear Damage, 21 May 1963, 1063 UNTS 265, as amended by the Protocol of 12 September 1997, Arts. II, IV.

¹³ Joint Protocol Relating to the Application of the Vienna Convention and the Paris Convention, 21 September 1988, 1672 UNTS 302.

¹⁴ Ibid, Arts. V, XI; Paris Convention on Third Party Liability in the Field of Nuclear Energy, 29 July 1960, 956 UNTS 251, as revised, Art. 3.

¹⁵ Convention on Supplementary Compensation for Nuclear Damage, 12 September 1997, IAEA Doc. INFCIRC/567, Arts. III–IV.

¹⁶ Civil Liability for Nuclear Damage Act, 2010 (India), s. 17(b).

¹⁷ Ibid, s. 46.

which India is a Party.¹⁸ The controversy has impeded implementation of the India-US Civil Nuclear Agreement and dampened commercial interest from American and French vendors. The Indian government has attempted to address supplier concerns through administrative arrangements, though their legal adequacy remains disputed.

Table 2: Comparative Overview of Nuclear Liability Regimes

Feature	Vienna Conv. (1997)	Paris Conv. (2004)	CSC (1997)	India CLNDA (2010)
Liability basis	Strict/Absolute	Strict/Absolute	Strict/Absolute	No-fault
Channelling	To operator	To operator	To operator	To operator (with recourse)
Liability limit	SDR 300 million	EUR 700 million	SDR 300 million (min.)	INR 1500 crore
Environmental damage	Covered	Covered	Covered	Covered
Supplier recourse	Not provided	Not provided	Not provided	Permitted (s. 17)
Geographic scope	EEZ included	EEZ included	EEZ included	National territory

NUCLEAR SECURITY LAW

A. The Convention on Physical Protection of Nuclear Material

¹⁸ P. Sands & P. Galizzi, 'The 1997 Vienna Convention on Civil Liability for Nuclear Damage' (1999) 48 International & Comparative Law Quarterly 475, 488.

The Convention on the Physical Protection of Nuclear Material (CPPNM), originally concluded in 1980 and significantly amended in 2005 (the Amendment entering into force in 2016), establishes binding obligations on States to protect nuclear material in international transport and, as amended, domestic use, storage, and transport.¹⁹ The 2005 Amendment was a direct response to the post-9/11 recognition that terrorist acquisition of nuclear or radiological material posed an existential threat. The Amendment extends CPPNM coverage to domestic nuclear facilities, requires States to enact legislation criminalising the sabotage of nuclear facilities, and establishes an obligation to share information relevant to the physical protection of nuclear material.

B. UN Security Council Resolution 1540 (2004)

Adopted under Chapter VII of the UN Charter, Security Council Resolution 1540 (2004) imposes legally binding obligations on all UN Member States to adopt and enforce effective measures against the proliferation of weapons of mass destruction and their means of delivery, with particular emphasis on preventing non-State actors from acquiring nuclear, biological, or chemical weapons.²⁰ Resolution 1540 requires States to enact domestic legislation criminalising the manufacture, acquisition, and transfer of such weapons and to adopt border controls and export regulations.²¹ The 1540 Committee monitors implementation.

C. Nuclear Security Summits and Soft Law

The Nuclear Security Summit (NSS) process, initiated by President Obama in 2010 and continuing through four summits until 2016, produced a substantial body of politically binding commitments - communiqués, national action plans, and gift baskets - that have shaped State practice significantly even without formal treaty character. The NSS process resulted in the removal of highly enriched uranium (HEU) from multiple States, accelerated ratification of the CPPNM Amendment, and strengthened bilateral nuclear security cooperation.²² The IAEA's Nuclear Security Series of guidance documents serves as the primary technical reference for States implementing their nuclear security obligations.

¹⁹ Convention on the Physical Protection of Nuclear Material, 3 March 1980, 1456 UNTS 125, as amended 2005, Art. 2A.

²⁰ UN Security Council Resolution 1540, 28 April 2004, S/RES/1540 (2004) para. 2.

²¹ *Ibid*, para. 3.

²² IAEA, 'Nuclear Security Series No. 13: Nuclear Security Recommendations on Physical Protection of Nuclear Material and Nuclear Facilities (INFCIRC/225/Revision 5)' (NSS No. 13, IAEA, Vienna, 2011).

RADIOACTIVE WASTE MANAGEMENT AND DECOMMISSIONING

A. The Joint Convention

The Joint Convention on the Safety of Spent Fuel Management and on the Safety of Radioactive Waste Management, which entered into force in 2001, is the first international legal instrument to address the safety of civilian radioactive waste and spent fuel on a global scale.²³ Unique in its structure, the Joint Convention operates through a peer review mechanism: Contracting Parties submit national reports on the measures taken to implement their obligations, which are then reviewed by other Contracting Parties at triennial review meetings. This non-adversarial peer review model has proven effective in promoting continuous improvement.

B. Prohibition of Ocean Dumping

The 1972 London Convention and its 1996 Protocol prohibit the dumping of radioactive waste at sea.²⁴ The 1993 amendment to the London Convention established a moratorium on sea disposal of low-level radioactive waste, effectively extending the prohibition to all radioactive material. The evolution from permitting to prohibiting marine disposal reflects both advances in waste management technology and a precautionary approach to marine environmental protection. Under UNCLOS Article 194, States are also required to prevent, reduce, and control pollution of the marine environment from all sources, including land-based radioactive contamination.²⁵

INDIA'S LEGAL POSITION IN THE INTERNATIONAL NUCLEAR ORDER

A. India Outside the NPT

India occupies an anomalous position in international nuclear law. Having conducted its first nuclear test in 1974 (Pokharan-I) and declared itself a nuclear-weapon State following the 1998 Pokharan-II tests, India has never been party to the NPT. As an NPT non-party, India is not subject to IAEA comprehensive safeguards requirements under Article III. However, India's civilian nuclear facilities are subject to facility-specific safeguards under a 2009 India-IAEA

²³ Joint Convention on the Safety of Spent Fuel Management and on the Safety of Radioactive Waste Management, 5 September 1997, 2153 UNTS 357, Art. 30.

²⁴ Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter, 29 December 1972, 1046 UNTS 120; 1996 Protocol thereto, Art. 4(1).

²⁵ UN Convention on the Law of the Sea, 10 December 1982, 1833 UNTS 397, Art. 194(3)(a).

Safeguards Agreement concluded pursuant to the India-specific exception granted by the Nuclear Suppliers Group in 2008.²⁶

B. The India-US Civil Nuclear Agreement (123 Agreement), 2008

The Agreement for Cooperation Between the Government of India and the Government of the United States of America Concerning Peaceful Uses of Nuclear Energy (123 Agreement), signed on 10 October 2008, represented a watershed in India's integration into the civil nuclear order. The Agreement was made possible by the Hyde Act (2006) in the United States,²⁷ which granted a waiver from standard US nuclear cooperation requirements, and by the NSG's grant of a State-specific exemption from its full-scope safeguards requirement.

Under the 123 Agreement, the United States and India agreed to cooperate in civil nuclear energy including nuclear fuel supply, reactor technology transfer, and reprocessing of US-obligated nuclear material under IAEA safeguards in a jointly agreed facility.²⁸ The Agreement also contains a right of return provision obliging India to return US-obligated nuclear material and equipment in the event of a nuclear test.²⁹ The Agreement has not been fully operationalised, primarily due to disagreements over the supplier liability provisions of India's CLNDA, 2010.³⁰

C. India as a Responsible Nuclear State

India has increasingly engaged with the international nuclear security architecture. India joined the International Convention for the Suppression of Acts of Nuclear Terrorism (2005),³¹ ratified the 2005 CPPNM Amendment (2017), and concluded the CSC (2016). India participates actively in IAEA technical cooperation programmes and contributes to the IAEA's nuclear security fund. India's nuclear doctrine of credible minimum deterrence and no-first-use, while not embedded in any international legal instrument, has been consistently articulated as a responsible nuclear posture. India's application for full membership of the NSG remains pending, contested primarily by China.

²⁶ IAEA, 'Agreement between the Government of India and the International Atomic Energy Agency for the Application of Safeguards to Civilian Nuclear Facilities' (INFCIRC/754, IAEA, Vienna, 2009).

²⁷ Henry J. Hyde United States-India Peaceful Atomic Energy Cooperation Act, Pub. L. No. 109-401, 120 Stat. 2726 (2006).

²⁸ Agreement for Cooperation Between the Government of India and the Government of the United States of America Concerning Peaceful Uses of Nuclear Energy, 10 October 2008 (123 Agreement), Art. 6.

²⁹ Ibid, Art. 14(4) (right of return clause).

³⁰ S.K. Singh, 'India's Civil Nuclear Programme: Law, Policy and Paradox' (2012) 52 Indian Journal of International Law 1, 14.

³¹ Convention for the Suppression of Acts of Nuclear Terrorism, 13 April 2005, 2445 UNTS 89.

D. India's Three-Stage Nuclear Programme: Legal Dimensions of the Fast Breeder Reactor

A defining feature of India's civil nuclear strategy is the three-stage nuclear programme, conceived by Dr. Homi J. Bhabha, which seeks to achieve long-term energy self-sufficiency through sequential exploitation of uranium and thorium resources. Stage I relies on Pressurised Heavy Water Reactors (PHWRs); Stage II deploys Fast Breeder Reactors (FBRs) using plutonium derived from Stage I operations; and Stage III envisages advanced thorium-based reactors. The legal significance of this programme lies in its intimate relationship with nuclear material production, reprocessing, and safeguards - all of which engage India's obligations under its IAEA Safeguards Agreement and the broader international non-proliferation architecture.

India's Prototype Fast Breeder Reactor (PFBR) at Kalpakkam, Tamil Nadu, achieved criticality on 6 April 2026.³² From an international law perspective, this milestone is significant because the PFBR is designated as a civilian facility under India's Separation Plan - the schedule of civilian and military nuclear facilities communicated to the IAEA pursuant to the 2008 NSG exemption and the 2009 India-IAEA Safeguards Agreement.³³ As a civilian facility, the PFBR is expected to be brought under IAEA facility-specific safeguards once it attains commercial operation, creating a direct legal obligation for India to declare the facility, accept Agency inspections, and maintain nuclear material accountancy.

The reprocessing of spent fuel from PHWRs to extract plutonium for the FBR programme is a legally sensitive activity. Reprocessing is a dual-use technology - it can produce plutonium for civilian reactor fuel or, with modification, weapons-grade material. Under the India-IAEA Safeguards Agreement, reprocessing of safeguarded nuclear material is subject to IAEA approval and verification. India's right to reprocess US-origin nuclear material is also addressed in the 123 Agreement, which provides for consultations on a dedicated reprocessing facility under IAEA safeguards.³⁴ The operationalisation of the PFBR will generate increased plutonium throughput and correspondingly heightened safeguards obligations.

The governance of India's nuclear programme also raises questions of legal accountability. The Department of Atomic Energy (DAE) reports directly to the Prime Minister's Office, and the nuclear sector operates with limited parliamentary oversight and judicial scrutiny. The Atomic

³² Vajiram & Ravi Mains Team, 'FBR in India: How FBR Strengthens India's Nuclear Energy Strategy', Vajiram & Ravi Current Affairs (9 April 2026) <<https://vajiramandravi.com/current-affairs/fbr-strengthens-indias-nuclear-energy-strategy/>> accessed 10 April 2026.

³³ IAEA, 'India's Separation Plan' (INFCIRC/731, IAEA, Vienna, 2008); INFCIRC/754 (n 28).

³⁴ 123 Agreement (n 30) Art. 6(iii); INFCIRC/754 (n 28) Art. 14.

Energy Act, 1962 concentrates regulatory and operational functions within the same institutional framework, an arrangement that has been criticised as falling short of the IAEA's Safety Requirements for independent nuclear regulatory authority.³⁵ The Convention on Nuclear Safety, to which India is not a Party, specifically requires the establishment of a regulatory body independent from entities promoting nuclear energy - a standard that India's present governance structure does not fully satisfy. Operationalisation of the PFBR will renew calls for legislative reform to separate regulatory from promotional functions in India's nuclear sector.

EMERGING CHALLENGES FOR INTERNATIONAL NUCLEAR LAW

A. Nuclear Energy and Climate Change

The Paris Agreement's ambition to limit global temperature increase to 1.5°C above pre-industrial levels has renewed interest in nuclear energy as a low-carbon electricity source.³⁶ The International Energy Agency and the IPCC have identified nuclear as a component of a decarbonised energy mix.³⁷ This nuclear renaissance raises important questions for international law: Does climate necessity alter the interpretation of Article IV of the NPT? Can climate finance instruments under the UN Framework Convention on Climate Change legitimately support civil nuclear programmes? How does the treaty regime adapt to novel reactor technologies, including small modular reactors (SMRs) and advanced reactor concepts that may require different safeguards approaches?

B. Small Modular Reactors and the Legal Framework

Small modular reactors (SMRs), defined as reactors with generating capacity below 300 MWe, are designed for factory fabrication and modular deployment.³⁸ Their characteristics - smaller footprint, lower upfront capital, potential for remote siting - raise novel regulatory and safeguards questions. Can IAEA safeguards designed for large light-water reactors be efficiently applied to hundreds of small reactors potentially deployed across developing countries? Who bears liability when a factory-fabricated reactor module supplied by a foreign vendor causes

³⁵ Atomic Energy Act, 1962 (India), s. 3; Convention on Nuclear Safety, 17 June 1994, 1963 UNTS 293, Art. 8(2).

³⁶ Paris Agreement under the United Nations Framework Convention on Climate Change, 12 December 2015, UN Doc. FCCC/CP/2015/L.9/Rev.1, Art. 2(1)(a).

³⁷ IEA, 'Net Zero by 2050: A Roadmap for the Global Energy Sector' (IEA, Paris, 2021) 103.

³⁸ IAEA, 'Advances in Small Modular Reactor Technology Developments' (IAEA, Vienna, 2022) 5.

nuclear damage in a host State? How do export control regimes apply to modular reactor components? These questions require proactive legal development.³⁹

C. Cybersecurity and Nuclear Facilities

The Stuxnet incident in 2010 demonstrated that industrial control systems at nuclear enrichment facilities could be compromised by sophisticated cyber weapons, exposing the intersection of cyberlaw and nuclear security. There is no specific international treaty addressing cyber attacks on nuclear facilities. The applicability of existing international humanitarian law principles - distinction, proportionality, precaution - to cyber operations against nuclear installations remains debated.⁴⁰ The Tallinn Manual 2.0 on the International Law Applicable to Cyber Operations addresses some of these questions in the context of the law of armed conflict but does not resolve peacetime obligations.⁴¹ The IAEA has issued non-binding Computer Security guidance (NSS No. 17-T) but the legal framework remains significantly underdeveloped.

D. Nuclear Disarmament and the Treaty on the Prohibition of Nuclear Weapons

The Treaty on the Prohibition of Nuclear Weapons (TPNW), adopted by 122 States in 2017 and in force since January 2021, prohibits the development, testing, production, stockpiling, use, and threat of use of nuclear weapons. Article 1(e) thereof prohibits assistance to any prohibited activity,⁴² raising questions about whether TPNW Parties may participate in certain forms of nuclear fuel cycle cooperation with NWS. All five NPT-recognised NWS, India, Pakistan, and Israel boycotted the TPNW negotiations.⁴³ The TPNW-NPT relationship, and the TPNW's compatibility with existing civil nuclear cooperation frameworks, constitutes a significant legal and political tension in the contemporary nuclear order.

E. Fast Breeder Reactors, Thorium Cycles, and Safeguards Adaptation

The advancement of fast breeder reactor technology and the prospective deployment of thorium-based fuel cycles present two of the most consequential legal challenges for international safeguards law in the coming decades. The standard IAEA Comprehensive

³⁹ M. Hibbs, 'The Future of Nuclear Power in East Asia' (Carnegie Endowment for International Peace, Washington DC, 2011) 44.

⁴⁰ T. Rid, 'Cyber War Will Not Take Place' (2012) 35 Journal of Strategic Studies 5, 18; IAEA, 'Computer Security for Nuclear Security' (NSS No. 17-T, IAEA, Vienna, 2021).

⁴¹ M.N. Schmitt (ed.), Tallinn Manual 2.0 on the International Law Applicable to Cyber Operations (Cambridge University Press, Cambridge, 2017) Rule 80.

⁴² Treaty on the Prohibition of Nuclear Weapons, 7 July 2017, UN Doc. A/CONF.229/2017/8, Art. 1(e).

⁴³ D.H. Joyner, 'The Legal Meaning and Implications of Article VI of the Non-Proliferation Treaty' (2011) 37 Yale Journal of International Law 259, 274.

Safeguards Agreement model (INFCIRC/153) and the verification techniques developed thereunder were designed primarily for light-water and heavy-water reactor technology. Fast breeder reactors involve continuous conversion of fertile material to fissile plutonium in the reactor blanket, a process that generates nuclear material accounting challenges distinct from those arising in conventional reactor operations.⁴⁴ The IAEA has acknowledged the need to develop safeguards approaches adapted to fast reactors, but no standardised legally operative methodology has yet been adopted.⁴⁵ This gap between legal obligation - the duty of non-diversion under Article III NPT - and technical verification capacity is a structural vulnerability in the present safeguards system.

India's PFBR at Kalpakkam, as it advances toward commercial operation under IAEA facility-specific safeguards, will be an early test case for the adequacy of existing verification law applied to an operating fast breeder reactor. The IAEA will need to negotiate a safeguards approach document with India specifying the inspection frequency, material balance areas, key measurement points, and accountancy procedures appropriate for the PFBR's fuel cycle. The legal principle of effective safeguards - requiring that verification measures provide timely detection of diversion of a significant quantity of nuclear material - must be operationalised through technically credible means for the fast reactor context.⁴⁶

The thorium fuel cycle raises a distinct legal gap. Thorium-232, as a fertile but non-fissile material, is currently not subject to IAEA safeguards obligations under INFCIRC/153, which applies to 'source material' (uranium and thorium) only when it is being processed for use in a nuclear reactor or after it has been irradiated.⁴⁷ Once irradiated, thorium-232 converts to uranium-233 - a fissile material with weapons-applicable characteristics - which does attract safeguards. However, the legal framework governing the accountancy, inspection, and verification of uranium-233 production in thorium reactors remains underdeveloped.⁴⁸ As India moves toward Stage III of its three-stage programme, the international community will face a pressing obligation to develop progressive law in this area, whether through amendment of safeguards agreements, new IAEA Board of Governors decisions, or a dedicated international instrument.

⁴⁴ INFCIRC/153 (n 8) paras. 74-78 (material balance areas and key measurement points).

⁴⁵ IAEA, 'Safeguards Approaches for Facilities: Fast Reactors' (STR-375, IAEA, Vienna, 2013) 3-9.

⁴⁶ Ibid, 12; INFCIRC/153 (n 8) para. 28 (timely detection standard: one significant quantity within conversion time).

⁴⁷ INFCIRC/153 (n 8) para. 34(c) (definition of source material including thorium).

⁴⁸ IAEA, 'Thorium Fuel Cycle - Potential Benefits and Challenges' (TECDOC-1450, IAEA, Vienna, 2005) 99-105.

CONCLUSION

The international legal framework governing civil nuclear energy has evolved substantially over seven decades, from the bare bones of the IAEA Statute and NPT to a dense web of treaty obligations, IAEA standards, export control guidelines, and soft-law commitments. This framework has demonstrably contributed to preventing the spread of nuclear weapons, promoting safety in nuclear operations, and establishing minimum standards of nuclear security globally.

Yet the framework faces mounting challenges. The fragmentation of the liability regime, exemplified by India's CLNDA departing from the standard channelling model, undermines the coherence of nuclear commerce. The non-universality of the NPT, the stalled disarmament pillar, and the absence of NWS from the TPNW create structural tensions. Emerging technologies - SMRs, advanced reactors, digital instrumentation - test the adaptability of verification and safety instruments developed for mid-twentieth-century reactor designs. Cyber vulnerabilities impose new threat vectors that existing legal instruments were not designed to address.

Progressive development of international nuclear law must address these lacunae through a combination of treaty amendment, new instruments where necessary, authoritative interpretation of existing obligations, and continued elaboration of IAEA standards and guidance. The universalisation of the Additional Protocol, the harmonisation of liability regimes through broader adherence to the CSC, and the negotiation of safeguards approaches adapted to SMR deployment deserve priority attention.

For emerging and re-emerging nuclear energy States, including India, the path forward lies in constructive engagement with the international nuclear legal order - not through seeking exception from its requirements but through contributing to its evolution. The inalienable right to peaceful nuclear energy, recognised in Article IV of the NPT and reflected in the foundational instruments surveyed here, carries with it an equal obligation of responsibility: to the international community, to future generations, and to the integrity of the non-proliferation regime upon which global nuclear order rests.